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BIORESOURCES

➤ DEFINITION AND EXAMPLES

- Bioresources are non-fossil biogenic resources which can be used by humans for multiple purposes, such as
 - ✓ To produce food
 - ✓ Substantial products, and / or
 - ✓ As energy carrier
- May include agriculture, forestry and biologically derived wastes
- Bioproducts such as antibiotics, single cell proteins, alcohol etc. are derived from bioresources
- Bioenergy source, such as biodiesel, syngas and biochar are derived from bioresources
- Bioresource applications provide eco-friendly environment and environmental sustainability
- May be divided into two major groups, such as natural bioresources and anthropogenic bioresources
- Natural bioresources are directly obtained from nature, like plants (biomass), animals (fats) and microbes (algae)
- Anthropogenic bioresources are originated through human activities, examples are rice husk, bagasse, wheat straw etc.

➤ CLASSIFICATION

May be classified as follows:

1) Primary Bioresource

2) Secondary Bioresource

3) Tertiary Bioresource

4) Quaternary Bioresource

- **Primary Bioresource**

- ✓ Generated for a specific purpose
- ✓ May be generated through forests, agriculture or aquaculture to enable the production of food, substantial products or eventually energy
- ✓ Examples are: grain, fish, potato, bamboo, algae, etc.

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- **Secondary Bioresources**
 - ✓ Generated during processing of the primary bioresources
 - ✓ Also obtained during industrial processing as by-product
 - ✓ Examples: molasses, bagasse, corncob, etc.
 - ✓ They are produced in large amount
 - ✓ They accrue genuine from vergin materials
 - ✓ They contain low amount of impurities
- **Tertiary Bioresources**
 - ✓ They are also parts from vergin materials
 - ✓ Separated along the processing chain
 - ✓ Compared to the secondary bioresources, they are residues
 - ✓ Occur rather in small amount at the generation place
 - ✓ Degradation during storage
 - ✓ They have lower value than secondary bioresources
- **Quaternary Bioresources**
 - ✓ Occur after a product was used
 - ✓ May be distinguished into short, mid and long term category
 - ✓ After begin of product use they are generated
 - ✓ In all cases of **feed and food consumption by humans and animals they are generated.**
 - ✓ Human feces and urine, animals excrements are the examples
 - ✓ Bioresources generated at short delay of food or feed are consumed at time delay of hours (animal feces, excretes,)
 - ✓ **In mid-term delay,** the quaternary bioresources are generated in days to months after **consumption. Packaging, newsprints materials** are the examples
 - ✓ **In long term delay,** bioresources may reach from years to centuries. Examples are: **wood construction materials, furniture m**aterials take decades to century till they become the wastes.

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➤ **MEDICINAL APPLICATIONS: INDIAN SUBCONTINENT PERSPECTIVE**

- The subcontinent is well known for its diversity of forest
- The age old healthcare traditions are generally known as Ayurveda/Hamdard
- As per one Sanskrit shloka, “There is no plant on the Earth, which doesn’t have medicinal properties.”
- As growth of the human population increased, natural area & resources were utilized to construct buildings.
- Such conditions have invited newer diseases.
- With the development in technology and industry, healthcare sector also grew rapidly.
- New generation of antibiotics are coming in the market.
- Such synthetic drugs are not digested easily in human body & have their side effect.
- Preparation of the synthetic drug may also involve air and water pollution.
- Nowadays people get attracted towards bioresource based medicines because they want to stay healthy & away from side effects.
- Ayurveda/Hamdard are science of life, which not only have curative drug but also have drugs which prevent the disease.
- This science include all natural resources, like herbal drugs.
- The uses of bio resources and their derivatives are best for human health.
- But the sources are limited.
- If we decide to use herbal drugs than we have to grow them.
- The ongoing growing recognition of medicinal plant is due to several reasons, including establishes faith of people on herbal medicine.
- Many NGOs have started cultivating medicinal plants & botanical garden.
- Herbs are natural products and their chemical composition varies depending on several factors and therefore varying from people to people, from energetic decoctions to the use of herbal extracts following Western methodologies of mainstream medicine.

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- **In every country traditional medicines find foundation in magical or religious beliefs, or popular experience and the World Health Organization is engaged to establish definitive guidelines for methodology of clinical research and the appraisal of effectiveness of traditional medicine.**
- **The sub-continent has ancient traditional science like Ayurveda and many cultural medicine too.**
- **So its need of the time to utilize these natural Bio-resources in health sector.**
- **The Sub-continent has the richest area of medicinal plants.**
- **There is an urgent need to establish the traditional bio-resource value in both national & international perspective.**

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BIORESOURCE TECHNOLOGY

➤ INTRODUCTION

- **Bioresource technology is an interdisciplinary area which integrates engineering design and biological sciences.**
- Here the engineering principles are applied for proper utilization, enhancement and sustainability of the world's natural resources (non-fossil and biogenic in nature) in a safe and profitable manner.
- It seeks solution of the societal problems through involvement of plants, animals, microbes and the environment.
- It is similar to biotechnology, except that it is based on agricultural and biological feedstocks.
- It is more general and encompasses a wider range of technologies and various elements, such as biomass, biological wastes treatment, biofuels and biorefineries, microbial products, physico-chemical and thermo-chemical processes for biomass and bioprocesses

➤ DETAILS FOR AREAS OF ACTIVITIES OF BIORESEARCH TECHNOLOGY

- **Biological Waste Treatment**
 - ✓ Activated sludge
 - ✓ Aerobic digestion
 - ✓ Anaerobic digestion
 - ✓ Biodegradation of toxic compounds
 - ✓ Biofiltration
 - ✓ Biological nutrients removal
 - ✓ Bioremediation
 - ✓ Biosorption of heavy metals
 - ✓ Environmental bioengineering
 - ✓ Phytoremediation of waste waste water
 - ✓ Biotreatment of food, dyes, and textile industries effluents
 - ✓ Solid waste management

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- **Biofuels and Biorefineries**
 - ✓ Biodiesel production
 - ✓ Bioethanol production
 - ✓ Biobutanol production
 - ✓ Biogas production
 - ✓ Biofuels from algae
 - ✓ Biohydrogen production
 - ✓ Energy and green house balances of biofuels
 - ✓ Life cycle assessment of biofuels
 - ✓ Microbial fuel cells
 - ✓ Modeling and design
- **Biomass as Feedstock**
 - ✓ **Enzymatic conversion of cellulose/lignocelluloses**
 - ✓ **Composting**
 - ✓ **Enzymatic pulping**
 - ✓ **Municipal waste**
 - ✓ **Vermicomposting**
- **Bioprocessing**
 - ✓ Algal utilization and harvesting, biocatalysis, bioleaching
 - ✓ Design and operation of bioreactors, heat and mass transfer effects, kinetics, modeling, optimization, protein engineering
 - ✓ Solid state fermentation, submerged fermentation
- **Microbial products**
 - ✓ Biofertilizers, biopesticides, biopolymers, biosurfactants, enzyme and other proteins, mushrooms, organic acids, metabolites,
- **Physicochemical and thermochemical processes for biomass**
 - ✓ Catalysis, combustion, gasification, pyrolysis, hydrothermal conversion, physical treatment, pretreatment for further processing, etc.